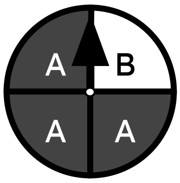
Answers

Worksheet

Problem solving with possible outcomes

Task A: Outcome tables and likelihood

1) Alex will spin this spinner twice.



spinner 1

a) Complete this outcome table to list all unique outcomes.

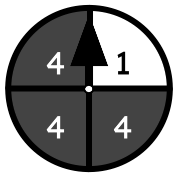
	A	B
A	AA	BA
B	AB	BB

b) Construct your own outcome table which shows the most likely outcome.

	A	A	A	B
A	AA	AA	AA	BA
A	AA	AA	AA	BA
A	AA	AA	AA	BA
B	AB	AB	AB	BB

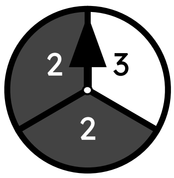
The outcome AA is the most likely, appearing 9 times.
Even if order didn't matter, AB only appears 6 times.

2) Aisha will spin spinner 2 and spinner 3 twice, and find the product of the numbers the spinners land on.



spinner 2

a) Complete this outcome table to list all unique outcomes.



spinner 3

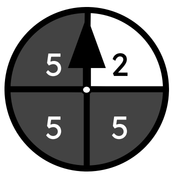
×	1	4
2	2	8
3	3	12

	1	4	4	4
2	2	8	8	8
2	2	8	8	8
3	3	12	12	12

b) Construct a second outcome table. Starting with the least likely outcome, rank the outcomes in order of likelihood.

Order of likelihood:
3 → 2 → 12 → 8

3) Sam and Lucas play a game.
Sam will spin spinner 4 twice and find the sum of the numbers the spinner lands on.
If the score is a double-digit number, Sam wins, if the score is either 8 or 9, it's a draw, and if the score is 7 or less, Lucas wins.



spinner 4

a) Construct the most efficient outcome table to show it is impossible for Sam and Lucas to draw.

	2	5
2	4	7
5	7	10

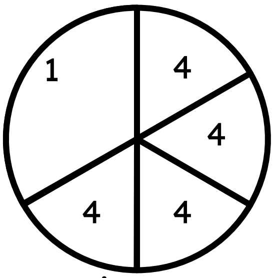
There are no outcomes that are either 8 or 9.

b) Construct the most efficient outcome table to find whether Sam or Lucas are more likely to win. Explain how you know.

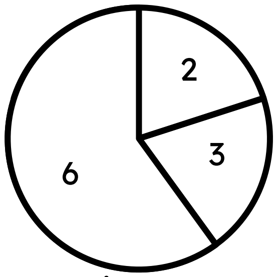
	2	5	5	5
2	4	7	7	7
5	7	10	10	10
5	7	10	10	10
5	7	10	10	10

Sam is more likely to win, as there are 9 double-digit outcomes, but only 7 outcomes that are 7 or less.

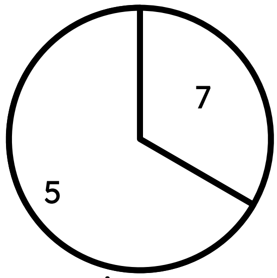
Task B: Outcomes and unequal likelihood



spinner A



spinner B

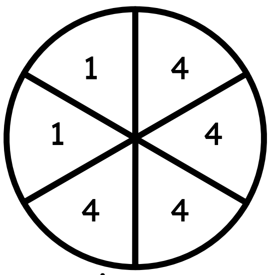


spinner C

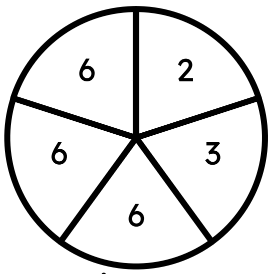
1) Jacob will spin spinners A and B once each, with the number each spinner lands on multiplied together. Construct an outcome table which lists each unique outcome to this trial.

	1	4
2	2	8
3	3	12
6	6	24

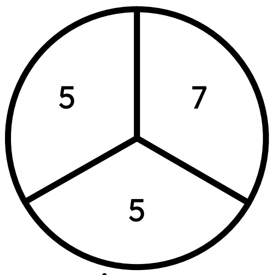
2) Split sectors on spinners A, B, C so that all outcomes on a spinner are equally likely.



spinner A



spinner B

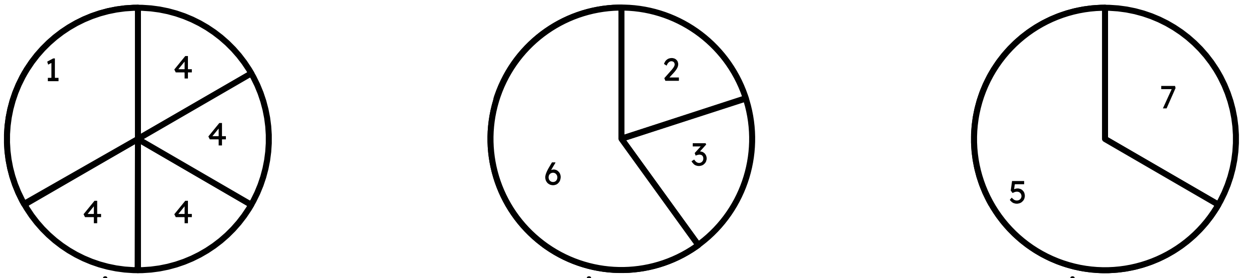


spinner C

3) Jacob will spin spinners B and C once each, with the number each spinner lands on multiplied together. Construct an outcome table which shows how likely each outcome is to occur.

(BC)	2	3	6	6	6
7	14	21	42	42	42
5	10	15	30	30	30
5	10	15	30	30	30

The outcome 30 appears 6 times out of 15 total outcomes.



spinner A spinner B spinner C

4) Jacob will spin spinners A and B once each, then spinners B and C once each, then spinners A and C once each. After each trial, the number each spinner lands on are multiplied together.

Which outcome is most likely to occur across all experiments?

(AB)	**2**	**3**	**6**	**6**	**6**
1	2	3	6	6	6
1	2	3	6	6	6
4	8	12	24	24	24
4	8	12	24	24	24
4	8	12	24	24	24
4	8	12	24	24	24

(AC)	**7**	**5**	**5**
1	7	5	5
1	7	5	5
4	28	20	20
4	28	20	20
4	28	20	20
4	28	20	20

20 is the most likely outcome to occur.

AB: The outcome 24 appears 12 times out of 30 total outcomes.

AC: The outcome 20 appears 8 times out of 18 total outcomes.

BC: The outcome 30 appears 6 times out of 15 total outcomes.